

ORBIT-NR

Optical-backhaul Regenerative Baseband In-orbit Transmitter for New Radio

PolarFire FPGA Design Contest 2026 - Track 3 (Splash Kit)

1. Submission summary

Item	Value
Contest	PolarFire FPGA Design Contest 2026 (Microchip)
Track	Track 3 - PolarFire Splash Kit
Project title	ORBIT-NR: Optical-backhaul Regenerative Baseband In-orbit Transmitter for NR. 5G NR NTN downlink OFDM transmitter for optically networked regenerative payloads
Team	Karthik Selvan (NLR - Royal Netherlands Aerospace Centre, NL); Patrice Rancé (Freelance engineer, FR)

2. Project Overview

Problem. 3GPP Release 17 and 18 define non-terrestrial networks (NTN) so that unmodified 5G-New Radio (NR) terminals connect through satellites. The next step, under study for Release 19, is the regenerative payload: the 5G base station next-generation Node B (gNB) physical layer runs on board and the satellite becomes a network node. On-board baseband must tolerate radiation, fit a small power budget and stay reconfigurable as the standard evolves. Published NR physical-layer implementations target SRAM-based FPGAs, GPUs or ASICs and meet none of these together. In parallel, Optical Inter-Satellite Links (OISL), such as the Optical Communication Terminals (OCT) developed by TNO and NLR, commercialized by FSO Instruments as Resolite portfolio will demonstrate on the Dutch MoD PAMI-1 (Platform for Autonomous Military Intelligence) mission, turn the space segment into a routed optical backhaul. Missing is the NR modem at the satellite edge connecting that backhaul to the RF user link.

Solution. We implement a 5G NR NTN downlink OFDM transmitter on the PolarFire Splash Kit. Transport blocks arrive over Gigabit Ethernet, representing the optical backhaul. A Mi-V RISC-V soft core runs the control plane (scheduling, numerology, DMRS, LDPC configuration). The RTL datapath implements CRC, NR LDPC encoding (BG1/BG2), rate matching, scrambling, QAM mapping, resource-element mapping, a 2048-point IFFT, cyclic prefix insertion and interpolation, on PolarFire math blocks and LSRAM. Baseband IQ leaves through one PolarFire transceiver lane in loopback, is captured in DDR4 and compared bit-exact against a MATLAB 5G Toolbox reference. The architecture is developed as an NR-compliant transmitter using a model-based flow (Simulink, HDL Coder) for DSP blocks and VHDL for control and interfaces.

Target application. Regenerative NTN payloads in optically networked constellations, with a migration path from MPF300T to radiation-tolerant RT PolarFire.

Expected outcome. A working, bit-exact NR downlink transmitter on the Splash Kit with published RTL, resource and timing figures and a reproducible verification flow. Phase 2 adds the receiver with NTN Doppler and timing synchronisation.

3. Innovation & Expected Impact

Novelty. To our knowledge no NR transmitter has been demonstrated on a flash-based FPGA. PolarFire configuration memory is immune to single-event upsets and its LSRAM carries SECDDED, so the same RTL moves to RT PolarFire without an architectural change. This is a first step towards a gNB physical layer that can fly on a small satellite at a few watts.

Key differentiators.

1. OISL-to-NR bridge. The transmitter is designed as the node between an optical inter-satellite backhaul and the RF user link, not as a stand-alone modem: packet ingest, framing and scheduling are built for that role.

2. Hardware/software co-design. Mi-V RISC-V owns the control plane; LDPC encoding, IFFT and mapping run as pipelined RTL on PolarFire math blocks. Numerology, bandwidth and IFFT size are runtime parameters, not synthesis constants.

3. Bit-exact verification. Every block is checked against a MATLAB 5G Toolbox reference with vectors that flow through Simulink, HDL Coder co-simulation and the Splash Kit loopback. Judges can reproduce the results from the published vectors.

4. Design evolution. Phase 2 adds the receiver with NTN-specific synchronisation (large Doppler, long round trip) and an exploratory study of an NLR-defined waveform on the same platform.

Impact. Regenerative NTN is on the roadmap of every major constellation and of 3GPP. A radiation-tolerant, reconfigurable NR baseband on a commercial low-power FPGA lowers the entry cost for small-satellite operators and for national programmes such as the Dutch PAMI series. RTL, test vectors and documentation will be released so others can build on them.

4. Development platform

Platform: PolarFire Splash Kit, MPF300T-1FCG484E. The kit has no ADC/DAC and no SFP+ cage; the board interfaces below are the ones the design uses.

Interface	Used for	Notes
RJ45 1 GbE (VSC8541, RGMII)	Transport block ingress from host	UDP/IP core in VHDL; the team has this core running on the Splash Kit
Transceiver lane 0 on FMC LPC	Baseband IQ output, loopback	Near-end PMA loopback first; external FMC loopback as option; up to 12.7 Gb/s
DDR4 x32	Capture buffer for any datapath tap	Libero DDR4 controller reference for the Splash Kit
USB-UART (FTDI)	Mi-V console, readback, debug	SoftConsole, SmartDebug
On-board FlashPro5	Programming and debug	-
PCIe x4 edge, SPI flash	Not used in phase 1	PCIe is a candidate host interface for phase 2

5. System Block Diagram (Submission item)

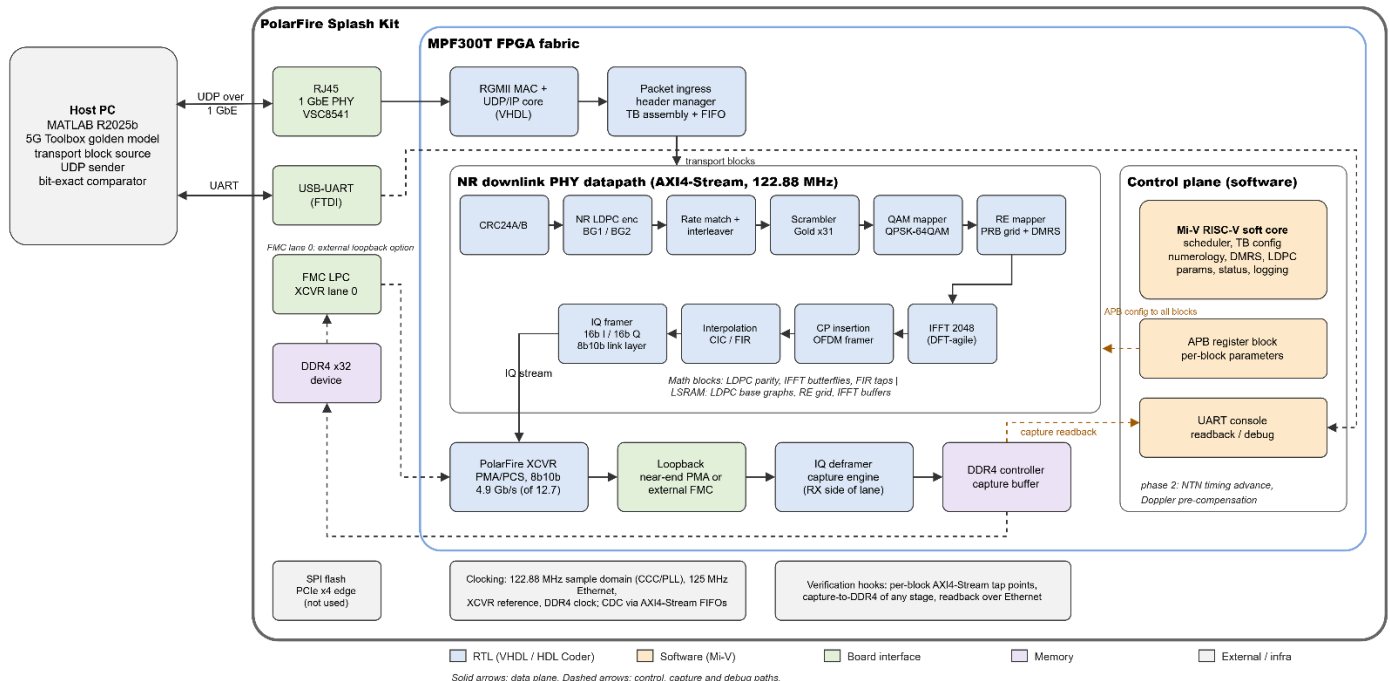


Figure 2. System block diagram - NR NTN downlink OFDM transmitter on the PolarFire Splash Kit

Data plane. Transport blocks are sent from the host over UDP on 1 GbE. The ingress block reassembles transport blocks from UDP payloads, checks sequence numbers and presents them on an AXI4-Stream interface. The NR downlink PHY datapath runs in a single 122.88 MHz clock domain and produces baseband IQ samples. The IQ framer packs 16-bit I and 16-bit Q into an 8b10b link layer on one PolarFire transceiver lane. The lane runs in loopback: near-

end PMA loopback inside the transceiver first, an external cable on the FMC connector as an option. The receive side of the lane deframes the samples and writes them into DDR4 through the capture engine. The host reads the captured samples back over Ethernet and compares them with the MATLAB reference.

Control plane. A Mi-V RISC-V soft core configures every block through an APB register block: MCS, transport block size, LDPC base graph and lifting size, redundancy version, RNTI, cell ID, PRB allocation, DMRS configuration, numerology and IFFT size. The same core drives the capture engine, so any AXI4-Stream tap point in the datapath can be routed to DDR4. A UART console is used for status and debug.

Clocking. 122.88 MHz sample-domain clock from a CCC/PLL, 125 MHz for Ethernet, the transceiver reference clock and the DDR4 clock. Clock domain crossings use AXI4-Stream FIFOs.

6. System context

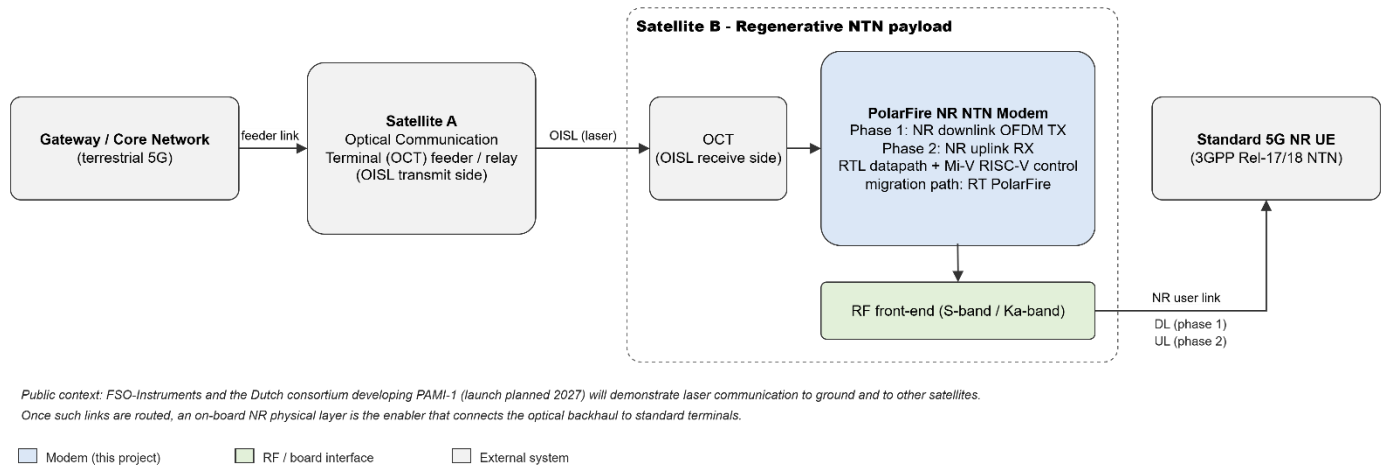


Figure 1. System context - NR NTN modem between an Optical Inter-Satellite backhaul and the RF user link

The transmitter is the downlink half of a regenerative NTN payload on a satellite that receives its data over an optical inter-satellite link. In the contest demonstrator the optical backhaul is represented by the 1 GbE ingress; the RF front end is represented by the transceiver loopback and capture. Nothing in the datapath depends on that substitution, which is what makes the migration to a flight architecture on RT PolarFire an engineering task rather than a redesign.

7. Transmitter datapath

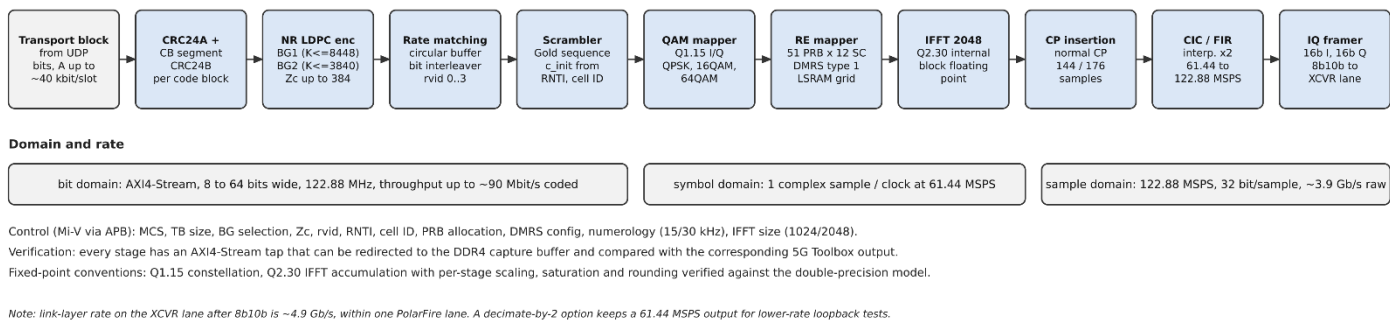


Figure 3. NR PDSCH transmitter datapath - data types and rates

Parameter	Phase 1 value	Runtime selectable
Channel	PDSCH with DMRS (type 1)	PRB allocation, DMRS config
Numerology	30 kHz SCS ($\mu = 1$)	15 kHz ($\mu = 0$)
Channel bandwidth	20 MHz, 51 PRB	10 MHz (24 PRB), 15 MHz (38 PRB)
IFFT size / sample rate	2048 / 61.44 MSPS	1024 / 30.72 MSPS

Parameter	Phase 1 value	Runtime selectable
Cyclic prefix	Normal (144, 176 samples)	-
Modulation	QPSK, 16QAM, 64QAM	MCS table index
Channel coding	NR LDPC BG1 and BG2, Zc up to 384	BG, Zc, rvid
Scrambling	Gold sequence, c_init from RNTI and cell ID	RNTI, cell ID
Output IQ format	16-bit I, 16-bit Q, 8b10b framed	x2 interpolation on/off
Peak coded throughput	~90 Mbit/s (64QAM, 51 PRB)	-

Fixed-point conventions follow the team's existing practice: Q1.15 for constellation points, Q2.30 accumulation inside the IFFT with per-stage scaling, saturation and rounding verified against the double-precision model. Word growth through the IFFT and the interpolation filter is budgeted from the Simulink fixed-point model before RTL is generated.

8. Hardware/software partition

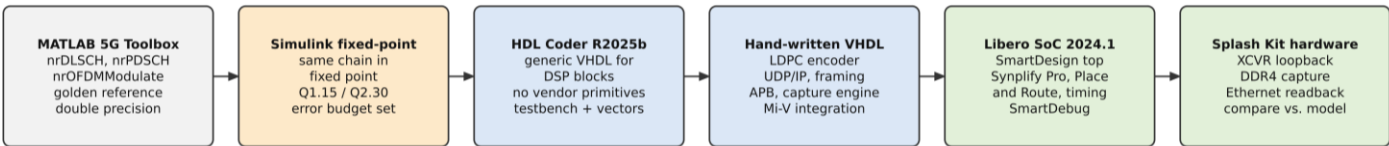
TBA

9. Resource and timing estimate

TBA

Target clock: 122.88 MHz for the datapath. PolarFire -1 speed grade math blocks and LSRAM support this with pipelined designs; the LDPC encoder is the block most likely to need attention, see risks.

10. Model-based design and verification



Verification ladder (each rung must pass before the next)

- V1 Block level: HDL Coder co-simulation, each block bit-exact against the Simulink fixed-point model.
- V2 Chain level: ModelSim simulation of the full VHDL chain with 5G Toolbox vectors; EVM of the IFFT output against the double model.
- V3 Hardware: same vectors sent over Ethernet, captured after loopback in DDR4, read back and compared sample by sample.
- V4 Standard conformance: captured waveform demodulated with nrPDSCHDecode in MATLAB; block error rate zero at the tested MCS set.
- V5 Performance: resource, timing and power report from Libero; sustained throughput measurement on hardware.

Where SmartHLS is used: candidate for the rate-matching bit-interleaver and DMRS sequence generation; decision after profiling the RTL alternative.
AI tools disclosure (rule 13): documentation and code review assistance; all RTL reviewed, simulated and validated on hardware by the team.

Figure 4. Model-based design and bit-exact verification ladder

11. Phase 2: design evolution

The final-stage judging rewards evolution beyond the proposal. Phase 2 is scoped now so that phase 1 decisions (register map, capture infrastructure, verification vectors) support it without rework.

- NR uplink OFDM receiver on the same platform: PUSCH de-mapping, LDPC decoding (layered min-sum on math blocks), channel estimation from DMRS.
- NTN-specific synchronisation: timing advance and Doppler pre-compensation from ephemeris, delivered by the Mi-V control plane; large frequency offset acquisition in the receiver.
- Exploratory study of an NLR-defined waveform for the optical inter-satellite feed, reusing the same ingest, framing and verification infrastructure.

12. References

1. Microchip, PolarFire FPGA Design Contest 2026, Official Rules and Terms and Conditions.
2. Microchip, UG0786 PolarFire FPGA Splash Kit User Guide (interfaces: PCIe Gen2 x4, 1 GbE, DDR4, FMC LPC with one transceiver lane, UART, SPI flash).
3. 3GPP TS 38.211, 38.212, 38.214; TR 38.821 (NTN solutions for NR); Release 17/18 NTN work items.
4. FSO Instruments, PAMI-1, <https://www.fso-instruments.nl/item/pami-1/>
5. Holland High Tech, PAMI-1 is the first in a series of satellites to strengthen Dutch space capabilities, <https://www.hollandhightech.nl/en/news-calendar/news/pami-1-is-the-first-in-a-series-of-satellites-to-strengthen-dutch-space-capabilities>
6. Netherlands Ministry of Defence, Nieuwe satelliet van Nederlandse makelij (22 Nov 2024), <https://www.defensie.nl/actueel/nieuws/2024/11/22/nieuwe-satelliet-van-nederlandse-makelij>
7. MathWorks, 5G Toolbox and HDL Coder R2025b documentation.